

CASEY DEL CARPIO BARTON

APPLIED AI SYSTEMS ARCHITECT | AGENT INFRASTRUCTURE
ENGINEER

Builds the operating layer that makes powerful AI dependable enough to use.

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69/69

Receipt Router tests

166 + 19

source + memory tests

148/148

Helix tests recorded

0

external flagship actions

SYSTEMS POSITIONING

Inspection discipline for AI systems.

Applied AI systems architect who builds the operating layer around model capability: task ownership, tool contracts, authority, evidence states, controlled failure, persistence, and completion receipts. Brings cross-domain discipline from software engineering, residential systems inspection, scientific measurement, field operations, and compressed-gas safety.

SELECTED EXECUTION

Proof that reveals engineering judgment.

TEST VERIFIED Portfolio Receipt Router Replaced unsupported autonomous-control semantics with a local, fail-closed evidence router while preserving compatibility. 69/69 tests; two CI paths; artifact 8910423397; zero external actions.	148/148 RECORDED Job Application Helix Evidence-governed hiring and portfolio orchestration across role research, source quality, claim calibration, package state, and follow-up. Recorded repository tests; current release and deployment gates remain distinct.
62/62 RECORDED Agent Coordinator Deterministic task ownership, dependency order, capacity, stable priority, shared budgets, and explicit refusal. Recorded Python tests; hosted promotion remains a separate state.	EXECUTION BLOCKED Microcode Governance Firmware manifests, drift, SBOMs, provenance, compatibility policy, and approval-gated rollout planning. Reviewed; private CI failed before runner execution; not represented as test verified.

EXPERIENCE

GlacierEQ - Founder / Applied AI Systems Builder

Jan 2025-Present

- Design and implement evidence-bound systems spanning agent orchestration, application intelligence, document and evidence pipelines, memory and continuity, infrastructure governance, and human-machine interfaces.
- Convert ambiguous architectures into typed contracts, runnable slices, deterministic tests, explicit refusal behavior, machine-readable facts, and reviewable completion receipts.
- Govern a multi-repository portfolio by separating source presence, review, execution, deployment, authority, and verified completion so recruiter, technical, and machine views cannot drift.
- Use AI-assisted development as an inspectable engineering process combining source review, bounded generation, adversarial checking, deterministic validation, and human judgment.

EXPERIENCE CONTINUED

Diamond Head Home Inspections - Certified Home Inspector

2020-2024

- Inspected residential systems and translated field observations into clear, decision-ready reports under time constraints.
- Applied disciplined evidence capture, uncertainty labeling, defect prioritization, interacting-system analysis, and plain-language communication.
- Developed the operating habit now used in software architecture: observe first, distinguish symptom from cause, test the boundary, and avoid unsupported conclusions.

Hi-Class Home Services / Hi Class Maintenance Oahu LLC - Owner-Operator, Building Systems and Field Services

2017-Present

- Scope residential repair and service work, prepare estimates, coordinate field execution, and communicate constraints and completion status with clients.
- Work across practical building-system problems including minor carpentry, plumbing, electrical fixture replacement, drywall, painting, flooring, and maintenance planning.
- Translate incomplete real-world requirements into bounded work packages, explicit assumptions, and usable closeout artifacts.

CROSS-DOMAIN FOUNDATION

A rare operating perspective.

SCIENTIFIC MEASUREMENT Sea-urchin morphometric research University of Hawaii, 2016-2017. Processed physical measurements of gill structures to support visual species identification.	COMPRESSED-GAS SYSTEMS Scuba Tank Technician Hawaiian Diving Adventures and UH Dive Office, 2016-2017. Tank filling, transport, inspection and repair support, and gas mixing under safety-critical procedures.	FIELD SYSTEMS Inspection and repair operations Practical judgment around interacting systems, uncertainty, failure consequences, client communication, and closeout quality.
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CAPABILITIES

Architecture Agent infrastructure, multi-agent coordination, application intelligence, authority and approval models, state machines, event design, idempotency, failure and recovery, human-in-the-loop systems.	Evidence and delivery Provenance, claim-to-source mapping, deterministic testing, JSON Schema, CI/CD, static analysis, dependency auditing, integrity hashing, release receipts, and machine-readable interfaces.	Primary technology Python, TypeScript, JavaScript, SQL, Bash, Node.js, React, Next.js, FastAPI, REST, JSON-RPC 2.0, MCP, GitHub Actions, Docker, and Vercel. Prior scientific computing: R and MATLAB.
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EDUCATION

University of Hawaii at Manoa	B.S., Marine Biology - 2016	AWS Cloud Institute	Cloud Application Developer program - 2025-2026, in progress
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Earlier technical certifications recorded in prior resumes: PSI Visual Cylinder Inspector, Eddy Current Technician, Valve Repair Technician, Oxygen Cleaning Cylinder Technician; NAUI Rescue and Master Diver; PADI Enriched Air Diver. Current status should be confirmed before role-specific use.

Evidence boundary. Independent GlacierEQ work and bounded technical exhibits. Test counts refer only to their stated repository and scope. No company affiliation, proprietary access, production use, customer impact, formal people-management experience, current certification status, or hardware validation is claimed without direct evidence.